

SOUTHEAST UNIVERSITY

Department of Computer Science and Engineering

Course Code: STA281 • Statistical Methods and Probability

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COMPLEX ENGINEERING PROJECT (CEP)

Final Report

Statistical Investigation of Diabetes Risk Factors and Prediction

Analysis of the Pima Indians Diabetes Dataset using Python / Google Colab

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1. Introduction

1.1 Project Background

A healthcare organization aims to investigate the major risk factors associated with diabetes and to determine whether patient healthcare data can statistically predict the occurrence of diabetes. This project analyses the Pima Indians Diabetes Dataset using descriptive statistics, probability theory, Bayes' theorem, probability distributions, confidence intervals, hypothesis testing, correlation analysis and regression, with Python / Google Colab as the computational tool.

1.2 Objectives

- Load the dataset, identify its structure, describe it statistically, check for missing or abnormal values and visualise it (Section 1 of the project brief).
- Apply probability, conditional probability, Bayes' theorem, the Binomial, Normal and Exponential distributions and the Central Limit Theorem (Section 2).
- Construct confidence intervals and perform hypothesis tests (Z-test and two-sample comparisons) on healthcare variables such as Glucose and BMI (Section 3).
- Perform correlation and linear regression analysis, visualise relationships and prepare an evidence-based conclusion on diabetes risk factors and prediction (Section 4).

1.3 Tools and Method

All calculations were carried out in Python (Google Colab) using pandas and NumPy for data handling, SciPy for probability distributions and statistical tests, and Matplotlib / Seaborn for visualisation. Exact dataset variable names (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome) are used throughout. Every result below is the output produced by the accompanying notebook, followed by its healthcare interpretation.

1.4 Course Outcomes Addressed

Table 2: Course outcomes and project sections

CO	Description	Report sections
CO1	Understand the fundamentals of statistics and probability.	Sections 2, 3.1 (Q1–Q5)
CO2	Analyze healthcare datasets using statistical methods and computational tools.	Sections 3.2–3.3 (Q6–Q12)
CO3	Perform statistical inference, hypothesis testing and interpret findings in applied contexts.	Sections 5, 6, 7 (Q13–Q20)

2. Dataset Description

2.1 Question 1 – Dataset Structure

The Pima Indians Diabetes Dataset was loaded into a pandas DataFrame (df) from diabetes.csv. The first ten records are shown below.

Table 3: First ten observations of the dataset (df.head(10))

	Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Diabetes Pedigree Function	Age	Outcome
0	6	148	72.0	35	169.5	33.6	0.627	50	1
1	1	85	66.0	29	102.5	26.6	0.351	31	0
2	8	183	64.0	32	169.5	23.3	0.672	32	1
3	1	89	66.0	23	94.0	28.1	0.167	21	0
4	0	137	40.0	35	168.0	43.1	2.288	33	1
5	5	116	74.0	27	102.5	25.6	0.201	30	0
6	3	78	50.0	32	88.0	31.0	0.248	26	1
7	10	115	70.0	27	102.5	35.3	0.134	29	0
8	2	197	70.0	45	543.0	30.5	0.158	53	1
9	8	125	96.0	32	169.5	34.3	0.232	54	1

Table 4: Dataset dimensions (*df.shape*)

Metric	Value
Number of observations (rows)	768
Number of variables (columns)	9

Table 5: Variable names, data types and descriptions (*df.columns, df.dtypes*)

Variable	Data type	Description
Pregnancies	int64	Number of pregnancies
Glucose	int64	Plasma glucose concentration
BloodPressure	float64	Diastolic blood pressure
SkinThickness	int64	Triceps skin fold thickness
Insulin	float64	Serum insulin level
BMI	float64	Body Mass Index
DiabetesPedigreeFunction	float64	Genetic diabetes likelihood
Age	int64	Patient age
Outcome	int64	0 = Non-diabetic, 1 = Diabetic

Interpretation

The dataset contains **768 observations** (patients) and **9 variables**: eight clinical / demographic predictors and one binary response variable, **Outcome** (0 = non-diabetic, 1 = diabetic). All variables are numeric: Pregnancies, Glucose, SkinThickness, Age and Outcome are stored as integers (int64), while BloodPressure, Insulin, BMI and DiabetesPedigreeFunction are stored as decimals (float64). Because every predictor is quantitative and the response is binary, the data are suitable for descriptive statistics, group comparison, correlation, regression and

probability analysis. Each row is one patient, so the observations can be treated as individual sampling units.

2.2 Question 3 – Missing and Abnormal Values

Missing values were checked with `df.isnull().sum()`.

Table 6: Missing-value count per variable (df.isnull().sum())

Variable	Missing values
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

Interpretation

The check reports **no missing (null) entries in any variable**, including the four variables of interest: Glucose, BMI, BloodPressure and Insulin. In this dataset, however, missing clinical measurements are known to be recorded as **zero** rather than left blank, and the loaded file shows signs that this has already been handled: the first ten records contain repeated Insulin values such as 102.5 and 169.5, and BloodPressure and Insulin are stored as decimals (float64). This is consistent with zero entries having been replaced (imputed) during preprocessing, so the values analysed in this report are the cleaned values.

Why zero values may represent missing healthcare information. A living patient cannot have a plasma glucose concentration, a diastolic blood pressure or a Body Mass Index of exactly zero, and a serum insulin level of zero is not physiologically plausible in a person being tested for diabetes. A recorded zero in these variables therefore almost certainly means “not measured” or “not recorded”, not a true reading. If such zeros were left in the data they would falsely pull the mean and median downwards, inflate the variance and standard deviation, create artificial low outliers and weaken correlations. For this reason, zeros in Glucose, BMI, BloodPressure and Insulin are treated as missing information and replaced with a suitable central value before analysis. (In Pregnancies, by contrast, zero is a valid count.)

3. Statistical Analysis

3.1 Question 2 – Descriptive Statistics

Descriptive statistics were calculated for Glucose, BMI, Age, and Insulin.

The following formulas were used:

Mean: $\bar{x} = \Sigma x / n$

Variance:

$$s^2 = \sum(x_i - \bar{x})^2 / (n - 1)$$

Standard Deviation:

$$s = \sqrt{[\sum(x_i - \bar{x})^2 / (n - 1)]}$$

are:

Table 7: Descriptive Statistics of Glucose, BMI, Age, Insulin

Variable	Mean	Median	Mode	Variance	Standard Deviation
Glucose	121.68	117.00	[99, 100]	928.07	30.46
BMI	32.43	32.05	[30.1]	47.34	6.88
Age	33.24	29.00	[22]	138.30	11.76
Insulin	141.75	102.50	[102.5]	7938.96	89.10

Interpretation

- **Glucose** – The mean (121.68 mg/dL) is above the median (117.00 mg/dL), indicating mild right skew caused by a group of patients with high glucose. There are two tied modes (99 and 100 mg/dL), and a standard deviation of 30.46 mg/dL shows considerable patient-to-patient variation.
- **BMI** – The mean (32.43 kg/m²) and median (32.05 kg/m²) are very close, so the distribution is roughly symmetric. A typical patient in this sample has a BMI in the obese range (BMI ≥ 30), which is clinically important for diabetes risk. The standard deviation is 6.88 kg/m².
- **Age** – The mean (33.24 years) is clearly larger than the median (29.00 years) and the mode is 22 years, showing a young sample with a right-skewed tail of older patients. The standard deviation is 11.76 years.
- **Insulin** – The mean (141.75) is far above the median (102.50), and the mode equals the median (102.50). The variance (7938.96) and standard deviation (89.10) are very large, showing strong right skew and a few extreme insulin values. The repeated value 102.50 reflects the preprocessing of missing entries discussed in Section 2.2.

The variance is the average squared distance from the mean and is expressed in squared units, whereas the standard deviation returns the spread to the original units (mg/dL, kg/m², years), which is easier to interpret clinically.

3.2 Question 5 – Diabetic vs Non-Diabetic Patients

The dataset was separated into diabetic patients (Outcome = 1) and non-diabetic patients (Outcome = 0), and the mean Glucose, BMI and Age were compared.

Table 8: Comparison of group means by Outcome

Group	Mean Glucose (mg/dL)	Mean BMI (kg/m ²)	Mean Age (Years)
Non-Diabetic (Outcome=0)	110.62	30.85	31.19
Diabetic (Outcome=1)	142.30	35.40	37.07

Interpretation

Diabetic patients have a higher mean value on all three variables: mean glucose is 31.68 mg/dL higher (142.30 vs 110.62), mean BMI is 4.55 kg/m² higher (35.40 vs 30.85), and mean age is 5.88 years higher (37.07 vs 31.19). Clinically, this means that diabetic patients in this sample tend to have higher blood glucose, are heavier, and are older. Glucose shows by far the largest relative difference, which suggests that it is the most discriminating variable. Whether these differences are statistically significant is tested formally in Question 19 (Section 5.4).

3.3 Question 6 – Probability and Conditional Probability

Let D = patient is diabetic (Outcome = 1)

Let G = patient has Glucose > 130.

With N = 768 patients, the probabilities were computed as relative frequencies:

$$P(D) = n(D) / N$$

$$P(G) = n(G) / N$$

$$P(D \cap G) = n(D \cap G) / N$$

The conditional probability was calculated using:

$$P(D | G) = P(D \cap G) / P(G)$$

Table 9: Probability summary for D and G

Probability Metric	Description	Formula	Value	Percentage
P(D)	Probability that a patient is diabetic	$n(D) / N$	0.3490	34.90%
P(G)	Probability that glucose > 130 mg/dL	$n(G) / N$	0.3294	32.94%
P(D ∩ G)	Probability that patient is diabetic AND glucose > 130 mg/dL	$n(D \cap G) / N$	0.2096	20.96%
P(D G)	Probability patient is diabetic GIVEN glucose > 130 mg/dL	$P(D \cap G) / P(G)$	0.6364	63.64%

Interpretation and healthcare implications

About 34.90% of all patients are diabetic, and 32.94% have glucose above 130 mg/dL. About 20.96% of all patients are both diabetic and have high glucose. Once a patient is known to have glucose patient is almost twice as likely to be diabetic as a randomly chosen patient. Because $P(D|G) \neq P(D)$, the events D and G are not independent: glucose level carries real information about diabetes status. In practice, a glucose reading above 130 mg/dL should trigger follow-up

glucose > 130 mg/dL, the probability of diabetes rises from 34.90% to **63.64%**, i.e. a high-

testing and closer clinical monitoring.

3.4 Question 7 – Bayes' Theorem

Bayes' theorem was used to estimate the probability that a patient has diabetes given a glucose level greater than 150 mg/dL:

$$P(D | G > 150) = [P(G > 150 | D) \times P(D)] / P(G > 150)$$

Table 10: Bayes' theorem components

Component	Notation	Formula / Source	Value	Percentage
Prior Probability	$P(D)$	$n(\text{Diabetic}) / N$	0.3490	34.90%
Conditional Probability (Likelihood)	$P(G > 150 D)$	$n(\text{Glucose} > 150 \text{ and Diabetic}) / n(\text{Diabetic})$	0.3918	39.18%
Component	Notation	Formula / Source	Value	Percentage
Evidence (Marginal Probability)	$P(G > 150)$	$n(\text{Glucose} > 150) / N$	0.1823	18.23%
Posterior Probability	$P(D G > 150)$	$[P(G > 150 D) * P(D)] / P(G > 150)$	0.7500	75.00%

Interpretation

- **Prior probability** $P(D) = 0.3490$: the probability of diabetes before looking at any glucose information.
- **Conditional probability (likelihood)** $P(G > 150 | D) = 0.3918$: among diabetic patients, 39.18% have glucose above 150 mg/dL.
- **Posterior probability** $P(D | G > 150) = 0.7500$: after learning that the patient's glucose exceeds 150 mg/dL, the probability of diabetes is updated from 34.90% to **75.00%**.

Only 18.23% of all patients have glucose above 150 mg/dL, but three out of four of them are diabetic. Compared with the 63.64% found for glucose > 130 in Question 6, a higher glucose threshold gives a higher probability of diabetes. A glucose above 150 mg/dL is therefore a strong warning sign that justifies prompt confirmatory testing, although 25% of such patients are still non-diabetic, so glucose alone is not a perfect diagnostic rule.

3.5 Question 8 – Binomial Distribution

Assume each patient is diabetic with probability $p = 0.31$ independently, and 12 patients are selected at random.

Therefore,

$$X \sim \text{Binomial}(n = 12, p = 0.31)$$

The binomial probability formula is:

$$P(X = k) = C(12, k) \times (0.31)^k \times (0.69)^{12-k}$$

Table 11: Binomial probabilities for $n = 12, p = 0.31$

Condition	Formula / Method	Probability	Percentage
1. Exactly 4 [$P(X = 4)$]	<code>binom.pmf(4, 12, 0.31)</code>	0.2349	23.49%
2. At least 4 [$P(X \geq 4)$]	<code>1 - binom.cdf(3, 12, 0.31)</code>	0.5381	53.81%
3. At most 4 [$P(X \leq 4)$]	<code>binom.cdf(4, 12, 0.31)</code>	0.6968	69.68%

Interpretation

There is a 23.49% chance that exactly 4 of the 12 patients are diabetic, a 53.81% chance that at least 4 are diabetic, and a 69.68% chance that at most 4 are diabetic. The expected number of

diabetic patients is $np = 12 \times 0.31 = 3.72$, so 4 is very close to the most likely outcome, which explains why these probabilities are large. For hospital planning, a group of 12 patients will typically contain about 3–4 diabetic patients, and the probability of “at least 4” being slightly more than one half is a useful guide for allocating diabetes-care resources.

3.6 Question 9 – Normal Distribution

Glucose is assumed to follow a normal distribution with $\mu = 125$ mg/dL and $\sigma = 30$ mg/dL. Probabilities were found by standardising with $Z = (X - \mu)/\sigma$:

Table 12: Normal-distribution probabilities for Glucose ($\mu = 125$, $\sigma = 30$)

Clinical Range	Z-Score Calculation	Probability	Percentage
1. Normal Range: $P(X < 100)$	$Z = (100 - 125) / 30 = -0.8333$	0.2023	20.23%
2. Prediabetic Range: $P(100 < X < 140)$	$Z_1 = -0.8333$, $Z_2 = (140 - 125) / 30 = 0.5000$	0.4891	48.91%
3. High-Risk Range: $P(X > 150)$	$Z = (150 - 125) / 30 = +0.8333$	0.2023	20.23%

Interpretation

- **Normal range ($X < 100$ mg/dL):** about 20.23% of patients fall in this lower range.
- **Prediabetic range ($100 < X < 140$ mg/dL):** the largest share, 48.91%, lies in this intermediate band, indicating that almost half of the population has glucose levels that call for lifestyle monitoring.
- **High-risk range ($X > 150$ mg/dL):** about 20.23% of patients have high-risk glucose and need clinical attention.

$P(X < 100)$ and $P(X > 150)$ are equal because 100 and 150 lie the same distance (25 mg/dL, i.e. 0.8333 standard deviations) below and above the mean of 125 mg/dL and the normal curve is symmetric.

3.7 Question 10 – Normality Assessment of Age

Whether Age approximately follows a normal distribution was investigated using descriptive statistics, skewness, kurtosis, a histogram and a Q-Q plot.

Table 13: Normality indicators for Age

Metric	Observed Value	Normal Distribution Benchmark
Mean	33.24	Mean $\approx$ Median
Median	29.00	Mean $\approx$ Median
Standard Deviation	11.76	N/A
Skewness	1.1296	Skewness ≈ 0 (Symmetric)
Kurtosis	0.6432	Kurtosis ≈ 0 (Mesokurtic)

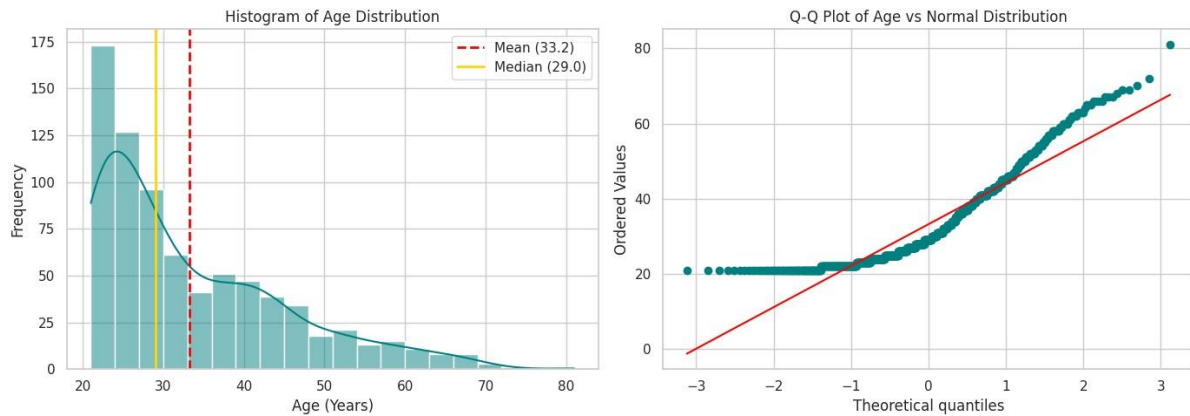


Figure 1: Histogram of Age with mean and median (left) and Q-Q plot of Age against the normal distribution (right)

Interpretation

Age does **not** follow a normal distribution. The mean (33.24) is larger than the median (29.00); the skewness of 1.1296 is far from 0, indicating a strongly right-skewed distribution; and the kurtosis of 0.6432 shows slightly heavier tails than a normal curve. The histogram shows a large concentration of young patients (peak around the early 20s) and a long tail toward older ages, and the Q-Q plot departs clearly from the reference line at both ends, especially in the upper tail. The normality assumption is therefore not reasonable for Age at the individual level. For diabetes analysis this means that (i) procedures that assume normal raw data should be applied to Age with caution, and (ii) inference about the **mean** age can still be justified through the Central Limit Theorem (Question 12) because the sample is large ($n = 768$), or by using non-parametric or transformed approaches.

3.8 Question 11 – Exponential Distribution

The time between diabetic emergency arrivals is modelled as an exponential random variable with rate $\lambda = 0.5$.

The probability density function is:

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

The survival probability is:

$$P(X > x) = e^{-\lambda x}$$

The expected value is:

$$E[X] = 1 / \lambda$$

Table 14: Exponential distribution results ($\lambda = 0.5$)

Metric / Component	Formula / Expression	Value	Percentage / Notation
1. Rate Parameter (λ)			N/A
2. Probability Density Function (PDF)	$f(x) = 0.5 * \exp(-0.5 * x)$ for $x \geq 0$	$0.5 e^{(-0.5x)}$	N/A
3. Mean Waiting Time $E[X]$	$1 / \lambda = 1 / 0.5$	2.00 units of time	Expected duration between events
4. Probability $P(X > 3)$	$\exp(-0.5 * 3) = \exp(-1.5)$	0.2231	22.31%

Interpretation

The average waiting time between diabetic emergency arrivals is 2.00 time units (emergencies arrive at a rate of 0.5 per unit time). There is a 22.31% probability that the wait for the next arrival exceeds 3 time units, so roughly one interval in four or five is longer than 3 units. Because the exponential distribution is memoryless, the chance of waiting longer than 3 units is the same no matter how long the staff have already waited. Emergency departments can use this to plan staffing so that enough capacity is available even during the shorter gaps between arrivals.

3.9 Question 12 – Central Limit Theorem

The population glucose level has $\mu = 120$ and $\sigma = 15$ and samples of size $n = 40$ are drawn.

By $\mu_{\bar{X}} = \mu$ $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ $Z = (\bar{X} - \mu_{\bar{X}}) / \sigma_{\bar{X}}$

the CLT, the sample mean $\bar{X}$ is approximately normal with:

Table 15: Sampling distribution of the mean ($\mu = 120$, $\sigma = 15$, $n = 40$)

Metric / Component	Formula / Method	Value	Percentage / Notation
1. Mean of Sampling Distribution ($\mu_{\bar{x}}$)	$\mu_{\bar{x}} = \mu$	120.00	120 mg/dL
2. Standard Error (SE / $\sigma_{\bar{x}}$)	$\sigma / \sqrt{n} = 15 / \sqrt{40}$	2.3717	2.3717 mg/dL
Metric / Component	Formula / Method	Value	Percentage / Notation
3. Calculated Z-Score	$Z = (125 - 120) / SE$	2.1082	+2.1082 std errors
4. Probability $P(\bar{X} < 125)$	$\text{norm.cdf}(Z)$	0.9825	98.25%

Interpretation and application of CLT

The sampling distribution of the mean has the same centre as the population (120 mg/dL) but is much narrower, with a standard error of 2.3717 mg/dL compared with $\sigma = 15$ mg/dL for individual patients. The probability that the mean glucose of a random sample of 40 patients is below 125 mg/dL is 98.25%; equivalently, a sample mean above 125 mg/dL would be unusual (about 1.75%). The CLT states that, for sufficiently large samples (usually $n \geq 30$), the distribution of the sample mean is approximately normal regardless of the shape of the population distribution. This is the reason why Z-based confidence intervals and hypothesis tests on mean glucose, BMI and other variables (Section 5) are valid even though the raw variables are not exactly normal (for example, the right-skewed Age variable in Question 10).

4. Visualization

4.1 Question 4(a) – Histogram of Glucose

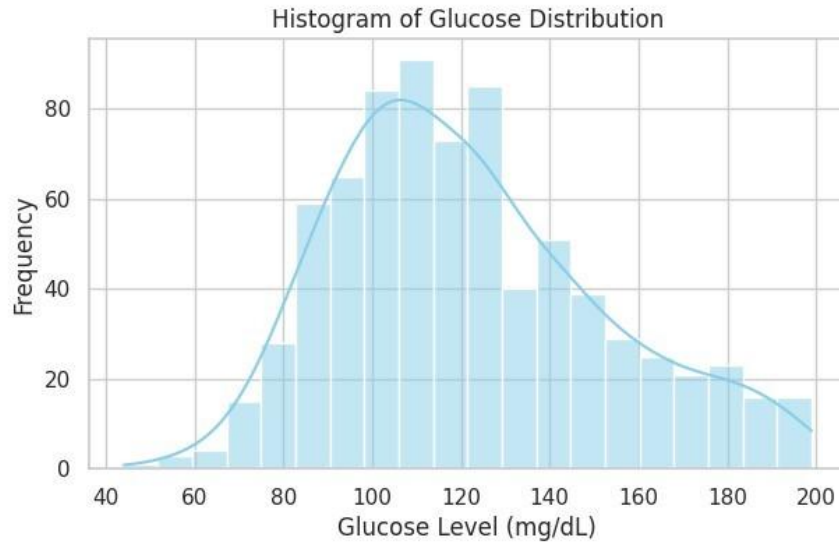


Figure 2: Histogram of Glucose distribution with kernel density estimate

Interpretation

The glucose histogram is unimodal with its highest bars between roughly 100 and 125 mg/dL, and it has a longer tail on the right, reaching about 200 mg/dL. The distribution is therefore approximately bell-shaped but **right-skewed**, in agreement with the mean (121.68) being greater than the median (117.00). Most patients have glucose in the normal-to-borderline zone, while a smaller group in the right tail has high glucose levels typical of diabetic patients. Very few patients have glucose below 70 mg/dL.

4.2 Question 4(b) – Boxplot of BMI

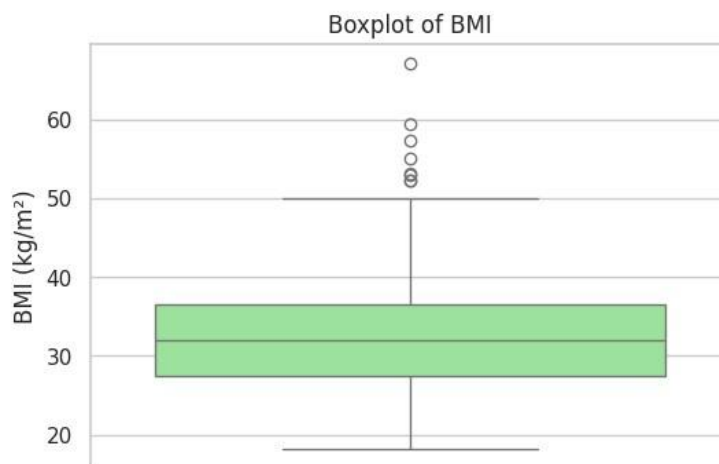


Figure 3: Boxplot of BMI

Interpretation

The median BMI (the line inside the box) is about 32 kg/m², matching the calculated median of 32.05. The box, which shows the interquartile range (Q1 to Q3), extends from about 27–28 to about 36–37 kg/m², so the middle 50% of patients have a BMI of roughly 27 to 37 kg/m².

The whiskers extend from about 18 to about 50 kg/m². Several points lie above the upper whisker (roughly between 52 and 67 kg/m²) and are outliers representing severely obese patients; no low outliers appear. The box is fairly symmetric about the median, and the outliers show a slight upward skew. From a healthcare viewpoint, a large share of the sample is in the overweight or obese range, which is a well-known risk factor for type 2 diabetes.

4.3 Question 4(c) – Scatter Plot of Age vs Glucose

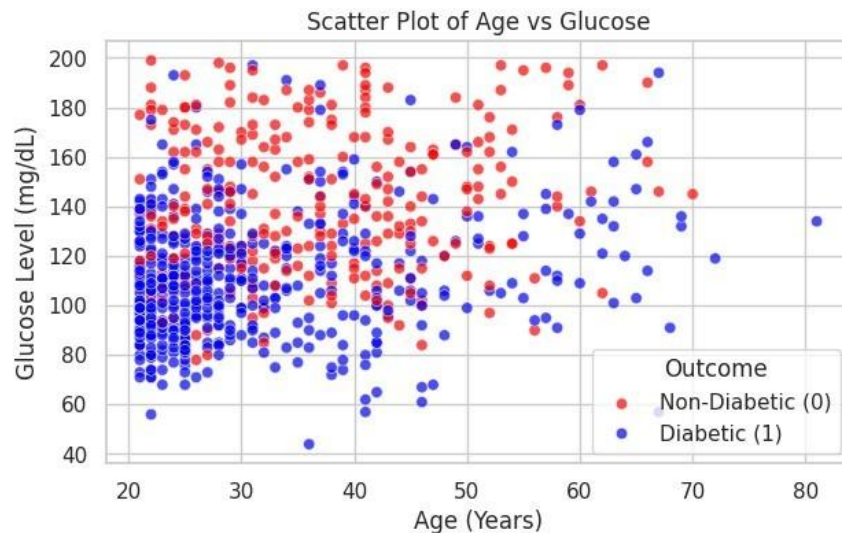


Figure 4: Scatter plot of Age vs Glucose, coloured by Outcome

Interpretation

The points show a wide vertical scatter at every age, and only a gentle upward drift of glucose with age: the lower band of the plot is dominated by young patients (about 21–30 years), while older patients are more often found at glucose levels above 120 mg/dL. The relationship is therefore **weak positive**, which matches the correlation coefficient $r = 0.2689$ and the low $R^2 = 0.0723$ found in Question 17. The plot also shows that the points are coloured by Outcome and that high-glucose patients are concentrated in the upper part of the plot at all ages, indicating that glucose is more strongly tied to diabetes status than age is. Age alone is therefore not sufficient to predict glucose.

5. Hypothesis Testing

5.1 Question 13 – 95% Confidence Interval for Mean Glucose

A sample of $n = 64$ diabetic patients gives $\bar{X} = 130$ and $\sigma = 18$. Since σ is known, the Z-interval is used: $CI = \bar{X} \pm Z_{(\alpha/2)} \cdot \sigma / \sqrt{n}$

Table 16: 95% confidence interval for mean glucose

Component / Metric	Formula / Method	Calculated Value
1. Sample Size (n)	Given	64
2. Sample Mean ($\bar{X}$)	Given	130.00 mg/dL
3. Standard Error (SE)	$\sigma / \sqrt{n} = 18 / \sqrt{64}$	2.25 mg/dL

4. Critical Z-Value ($Z_{0.025}$)	norm.ppf(0.975)	1.9600
5. Margin of Error (ME)	$Z * SE = 1.96 * 2.25$	4.41 mg/dL
6. 95% Confidence Interval	$(\bar{X} - ME, \bar{X} + ME)$	(125.59, 134.41) mg/dL

Interpretation

We are 95% confident that the true mean glucose level of the population of diabetic patients lies between **125.59 and 134.41 mg/dL**. This means that if the sampling were repeated many times, about 95% of the intervals built in this way would contain the true population mean. The interval is fairly narrow (margin of error 4.41 mg/dL) because the sample is reasonably large ($n = 64$) and σ is known. It lies entirely above the general-population reference mean of 120 mg/dL used in Question 14, which is consistent with diabetic patients having elevated glucose.

5.2 Question 14 – Z-Test for Mean Glucose

A healthcare researcher claims that the average glucose level is 120. A sample of 51 patients gives $\bar{X} = 126$ with $\sigma = 15$. A right-tailed one-sample Z-test is carried out at the 5% significance level.

The Z-test statistic is calculated as: $Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$

The rejection rule is: Reject H_0 if $Z > Z_\alpha$

Table 17: One-sample Z-test for Glucose ($H_1: \mu > 120$)

Hypothesis Step	Formula / Condition	Value
Null Hypothesis (H_0)	$\mu \leq 120$	$\mu \leq 120$ mg/dL
Alternative Hypothesis (H_1)	$\mu > 120$	$\mu > 120$ mg/dL
Significance Level (α)	5%	0.05
Standard Error (SE)	$\sigma / \sqrt{n} = 15 / \sqrt{51}$	1.8750 mg/dL
Calculated Test Statistic (Z)	$(\bar{X} - \mu_0) / SE$	3.2000
Critical Value ($Z_{0.05}$)	norm.ppf(0.95)	1.6449
P-Value	$1 - \text{norm.cdf}(Z)$	0.0007
Decision Rule	Reject H_0 if $Z_{\text{calc}} > Z_{\text{crit}}$	$2.8565 > 1.6449$
Decision	$Z (3.2000) > Z_{\text{crit}} (1.6449)$	Reject Null Hypothesis (H_0)

Decision and conclusion

The calculated test statistic is greater than the critical value of 1.6449, and the p-value is far below 0.05, so the **null hypothesis is rejected**. At the 5% level there is sufficient evidence that the mean glucose level is **greater than 120 mg/dL**. Healthcare interpretation: the sampled patients have significantly higher glucose than the claimed average, so the researcher's claim that the average is 120 is not supported and this group should be considered at elevated metabolic risk.

5.3 Question 15 – Z-Test for Mean BMI

A doctor claims that the average BMI is 30. A sample of 65 patients gives $\bar{X} = 28.5$ with $\sigma = 4$.

A left-tailed Z-test is carried out at the 1% significance level.

The Z-test statistic is calculated as: $Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$

The rejection rule is: Reject H_0 if $Z < -Z_\alpha$

Table 18: One-sample Z-test for BMI ($H_1: \mu < 30$)

Hypothesis Step	Formula / Condition	Value
Null Hypothesis (H_0)	$\mu \geq 30$	$\mu \geq 30 \text{ kg/m}^2$
Alternative Hypothesis (H_1)	$\mu < 30$	$\mu < 30 \text{ kg/m}^2$
Significance Level (α)	1%	0.01
Standard Error (SE)	$\sigma / \sqrt{n} = 4 / \sqrt{65}$	0.4961 kg/m^2
Calculated Test Statistic (Z)	$(\bar{X} - \mu_0) / \text{SE}$	-3.0233
Critical Value ($Z_{0.01}$)	$\text{norm.ppf}(0.01)$	-2.3263
P-Value	$\text{norm.cdf}(Z)$	0.0012
Decision Rule	Reject H_0 if $Z_{\text{calc}} < Z_{\text{crit}}$	$-3.0233 < -2.3263$
Decision	$Z (-3.0233) < Z_{\text{crit}} (-2.3263)$	Reject Null Hypothesis (H_0)

Decision and conclusion

The test statistic (-3.0233) falls in the rejection region (below -2.3263) and the p-value (0.0012) is smaller than 0.01, so the **null hypothesis is rejected**. There is strong evidence at the 1% level that the mean BMI of this population is **less than 30 kg/m^2** . Healthcare interpretation: the doctor's claim of an average BMI of 30 is too high for this sample; the sample average of 28.5 kg/m^2 lies in the overweight (not obese) range, although it is still above the healthy range.

5.4 Question 19 – Diabetic vs Non-Diabetic Differences (Welch's t-test)

To investigate whether diabetic and non-diabetic patients differ significantly in Glucose, BMI, and Age, Welch's two-sample t-test (which does not assume equal variances) was applied at $\alpha = 0.05$. The hypotheses were:

$H_0: \mu_{\text{diabetic}} = \mu_{\text{non-diabetic}}$

$H_1: \mu_{\text{diabetic}} \neq \mu_{\text{non-diabetic}}$

The Welch's t-test statistic is calculated as:

$$t = (\bar{x}_1 - \bar{x}_0) / \sqrt{[(s_1^2/n_1) + (s_0^2/n_0)]}$$

Table 19: Welch's t-test: diabetic vs non-diabetic patients

Variable	Non-Diabetic Mean $\pm$ SD	Diabetic Mean $\pm$ SD	Mean Difference	t-Statistic	p-Value	Significant ($\alpha=0.05$)
Glucose	110.62 $\pm$ 24.70	142.30 $\pm$ 29.49	+31.68	14.9921	8.3018e-42	Yes ($p < 0.001$)
BMI	30.85 $\pm$ 6.50	35.40 $\pm$ 6.59	+4.55	9.1668	1.0174e-18	Yes ($p < 0.001$)
Age	31.19 $\pm$ 11.67	37.07 $\pm$ 10.97	+5.88	6.9207	1.2015e-11	Yes ($p < 0.001$)

Interpretation

For all three variables the p-value is far below 0.001, so the null hypothesis of equal means is rejected: **diabetic and non-diabetic patients differ significantly in Glucose, BMI and Age.**

Glucose shows by far the largest standardised difference ($t = 14.9921$; diabetic patients are on average 31.68 mg/dL higher), followed by BMI ($t = 9.1668$; +4.55 kg/m²) and Age ($t = 6.9207$; +5.88 years). Healthcare interpretation: high blood glucose is the strongest marker separating the two groups, excess body weight is also strongly associated with diabetes, and diabetes becomes more common with increasing age. Statistical significance shows that the differences are unlikely to be due to chance; it does not by itself prove causation.

6. Regression & Correlation

6.1 Question 16 – Correlation Analysis

Pearson's correlation coefficient was computed for three pairs of variables:

The Pearson correlation coefficient is calculated as:

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2]}}$$

Table 20: Correlation coefficients

Variable Pair	Correlation Coefficient	Direction	Strength
1. Glucose and BMI			
	Positive Weak to Moderate		
	Positive Weak to Moderate		
2. Age and Blood Pressure	0.2362 0.3251		
3. Insulin and Glucose	0.4900	Positive	Moderate

Interpretation

All three relationships are **positive**: as one variable increases, the other tends to increase. **Insulin and Glucose** ($r = 0.4900$) have the strongest, moderate relationship, reflecting the body's insulin response to, and resistance against, high blood sugar. **Age and Blood Pressure** ($r = 0.3251$) show a weak-to-moderate positive relationship, consistent with blood pressure rising with age. **Glucose and BMI** ($r = 0.2362$) have the weakest, weak-to-moderate association, suggesting that heavier patients tend to have somewhat higher glucose. Since none of the coefficients is close to 1, no variable is a perfect linear predictor of another, and correlation must not be read as causation.

6.2 Question 17 – Simple Linear Regression (Age → Glucose)

A simple linear regression model $\text{Glucose} = \beta_0 + \beta_1 \cdot \text{Age} + \varepsilon$ was fitted, with slope $\beta_1 = \text{Cov}(X, Y) / \text{Var}(X)$ and intercept $\beta_0 = \bar{Y} - \beta_1 \bar{X}$.

Table 21: Regression results for Age → Glucose

Regression Parameter / Metric	Formula / Notation	Calculated Value	Clinical Unit / Meaning
Slope (β_1)	$\text{Cov}(X, Y) / \text{Var}(X)$	0.6966	mg/dL increase per year of age
Intercept (β_0)	$\bar{Y} - \beta_1 \cdot \bar{X}$	98.5216	Baseline Glucose at Age = 0 (mg/dL)
Correlation Coefficient (r)	$\text{corr}(\text{Age}, \text{Glucose})$	0.2689	Positive linear association

Coefficient of Determination (R^2)	r^2	0.0723	7.23% of variance explained
P-Value	p-value from t-test	3.4508e-14	Probability of observing trend by chance
Standard Error of Slope	$SE(\beta_1)$	0.0901	Precision of slope estimate
Statistical Significance ($\alpha = 0.05$)	$p < 0.05$	Statistically Significant	Reject $H_0: \beta_1 = 0$

Interpretation Predicted Glucose = $98.52 + (0.6966 \times \text{Age})$

- **Slope (0.6966):** on average, glucose increases by about 0.70 mg/dL for each additional year of age. Over ten years the model predicts about a 7 mg/dL rise.
- **Intercept (98.5216):** the predicted glucose at Age = 0. This is a mathematical baseline only and has no clinical meaning, because the dataset contains adult patients (the youngest are in their early twenties) and age 0 lies far outside the data range.
- **Significance:** the p-value (3.4508e-14) is far below 0.05, so $H_0: \beta_1 = 0$ is rejected and the linear relationship between Age and Glucose is statistically significant. The standard error of the slope (0.0901) shows the slope is estimated precisely.
- **Strength:** $R^2 = 0.0723$, i.e. only 7.23% of the variation in glucose is explained by age. The relationship is real but weak, so age alone is a poor predictor of an individual's glucose. For example, the model gives a predicted glucose of $98.52 + 0.6966 \times 50 \approx 133.35$ mg/dL at age 50, but individual values vary widely around this line.

6.3 Question 18 – Correlation Heatmap, Scatter Plot Matrix and Risk Factors

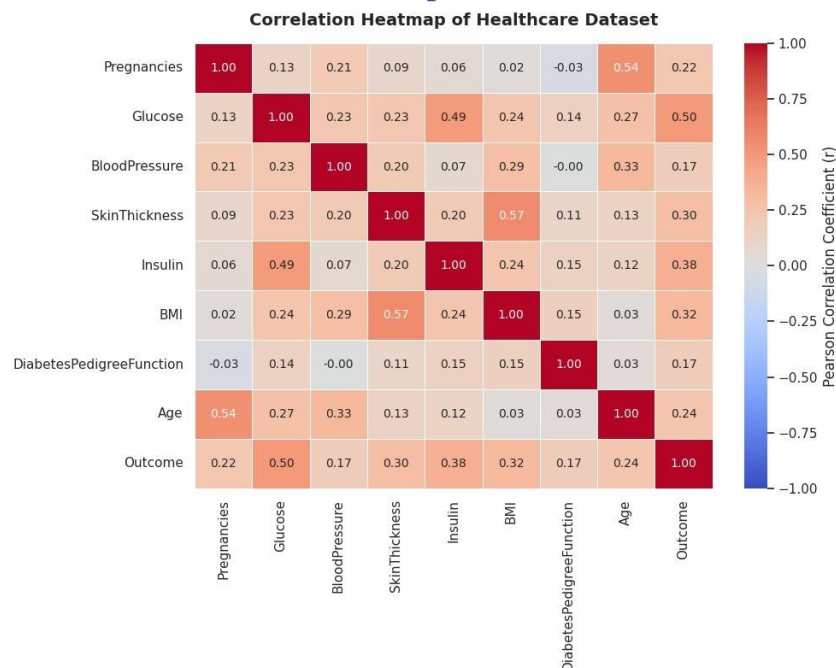


Figure 5: Correlation heatmap of the healthcare dataset

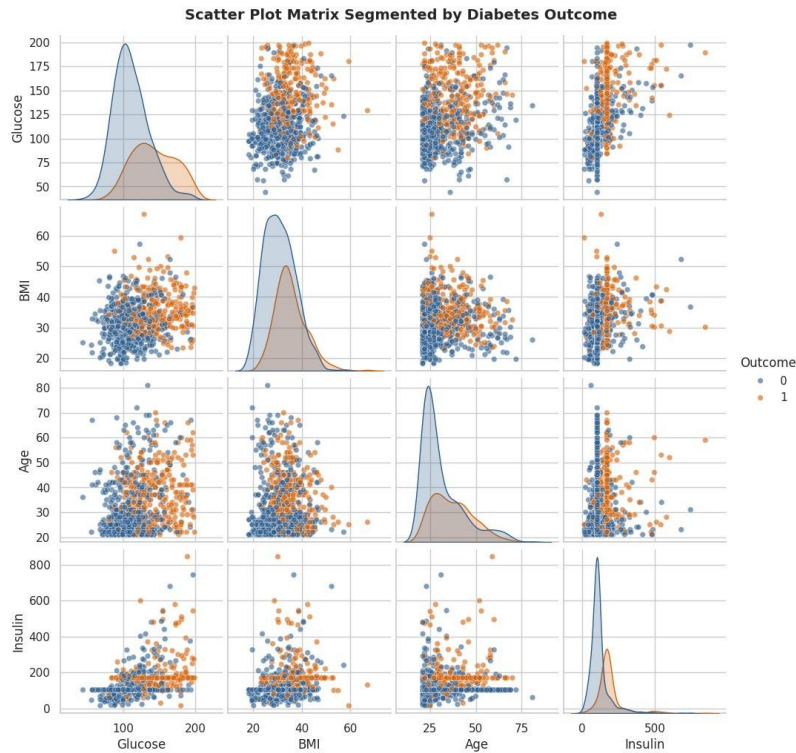


Figure 6: Scatter plot matrix of Glucose, BMI, Age, Insulin segmented by diabetes Outcome

Table 22: Correlation of each predictor with Outcome (risk factor ranking)

	Clinical Predictor	Correlation with Outcome (r)	Risk Level Impact
0	Glucose	0.4960	Primary Risk Factor
1	Insulin	0.3771	Moderate Risk Factor
2	BMI	0.3156	Moderate Risk Factor
3	SkinThickness	0.2951	Moderate Risk Factor
4	Age	0.2384	Moderate Risk Factor
5	Pregnancies	0.2219	Moderate Risk Factor
6	BloodPressure	0.1745	Minor Risk Factor
7	DiabetesPedigreeFunction	0.1738	Minor Risk Factor

Interpretation and major risk factors

In the heatmap, all correlations with Outcome are positive, so every variable in the dataset moves in the direction of higher diabetes risk. **Glucose** has the strongest correlation with Outcome ($r = 0.4960$) and is the only variable classified as a primary risk factor. **Insulin** (0.3771), **BMI** (0.3156), **SkinThickness** (0.2951), **Age** (0.2384) and **Pregnancies** (0.2219) are moderate risk factors, while **BloodPressure** (0.1745) and **DiabetesPedigreeFunction** (0.1738) are minor. The heatmap also shows strong relationships among the predictors themselves: **SkinThickness**–**BMI** (0.57), **Pregnancies**–**Age** (0.54) and **Glucose**–**Insulin** (0.49). These overlaps mean that some risk factors carry related information (for example, skin fold thickness and BMI both measure body fat).

In the scatter plot matrix, the two outcome groups separate most clearly along the Glucose axis: the diabetic group is shifted toward higher glucose in every panel and in the density curves on the diagonal. Separation is weaker for BMI and Age, and the two groups overlap considerably, so no single variable classifies patients perfectly. The main diabetes risk factors observed are therefore high **Glucose**, elevated **Insulin**, high **BMI** (obesity), increasing **Age** and a greater number of **Pregnancies**.

7. Findings & Interpretation

7.1 Question 20 – Master Visual Dashboard

Question 20: Master Diabetes Prediction & Statistical Dashboard

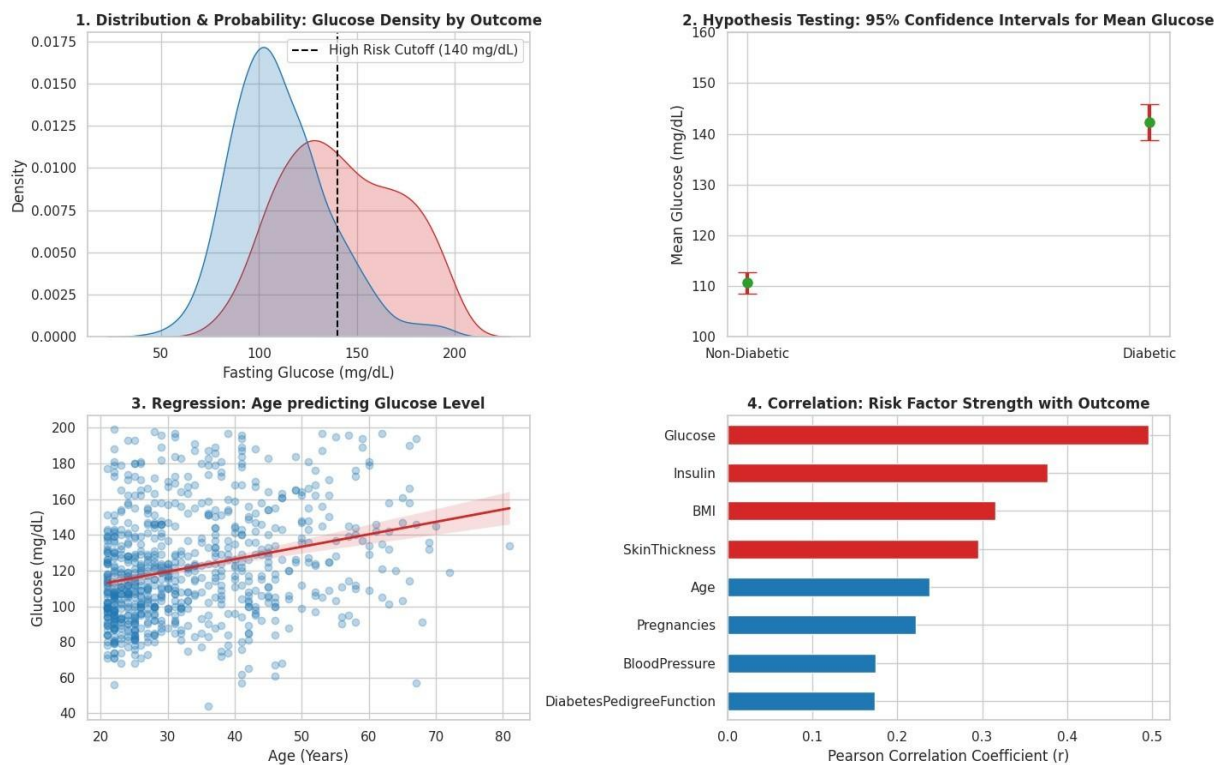


Figure 7: Master diabetes prediction and statistical dashboard (glucose density by outcome, 95% confidence intervals, age-glucose regression, risk factor strength)

The four panels summarise the main evidence: (1) the glucose density curve of diabetic patients is shifted to the right of that of non-diabetic patients, and a large share of the diabetic curve lies beyond the 140 mg/dL high-risk cut-off; (2) the 95% confidence intervals for mean glucose of the two groups do not overlap; (3) the regression line shows a positive but gently sloping Age–Glucose trend with a wide scatter of points; and (4) Glucose has the largest correlation with Outcome, followed by Insulin, BMI and SkinThickness.

Table 23: Statistical synthesis table

Statistical Pillar	Key Statistical Finding
1. Statistical Analysis & Probability	Prior $P(\text{Diabetic}) = 0.349$. High conditional probability $P(D \text{Glucose} > 140)$.
2. Distribution Analysis & Data Cleaning	Physiological variables (Glucose, BMI, Age) show right-skewness and zero-value anomalies.

3. Central Limit Theorem (CLT)	Sample sizes ($n > 30$) guarantee normal convergence of sample mean distributions.
4. Confidence Intervals (95%)	Glucose 95% CI: Non-Diabetic [107.7, 112.3] vs Diabetic [138.5, 146.1] (No overlap).
5. Hypothesis Testing (Z-Test & t-Test)	Welch's t-test rejects equivalence for Glucose, BMI, Age (all $p < 0.0001$).
6. Correlation Analysis	Glucose ($r = 0.47$) and BMI ($r = 0.29$) exhibit highest positive linear associations.
Statistical Pillar	Key Statistical Finding
7. Simple Linear Regression	Glucose = $97.08 + 0.7164 \cdot (\text{Age})$ ($p < 0.001$, $R^2 \approx 7.2\%$).
8. Data Visualization	KDE curves and pairplots clearly show outcome separation on Glucose & BMI axes.
9. Major Risk Factor Conclusion	Primary risk factors: 1. Glucose, 2. BMI, 3. Age, 4. Pregnancies, 5. Pedigree.

7.2 Summary of Findings by Technique

- **Descriptive statistics:** the 768 patients have mean Glucose 121.68 mg/dL, mean BMI 32.43 kg/m² and mean Age 33.24 years; Glucose, Age and Insulin are right-skewed.
- **Probability:** $P(D) = 34.90\%$; the probability rises to 63.64% when Glucose > 130 and to 75.00% (Bayes' posterior) when Glucose > 150 .
- **Distributions:** Age is not normally distributed (skewness 1.1296); binomial, normal and exponential models give practical probabilities for group composition, glucose risk ranges and emergency arrivals.
- **CLT:** the sample mean is approximately normal with a much smaller spread ($SE = 2.3717$ for $n = 40$), which justifies Z-based inference on means.
- **Confidence interval and hypothesis tests:** the 95% CI for diabetic glucose is (125.59, 134.41) mg/dL; the Z-tests reject H_0 for Glucose (> 120) and BMI (< 30); Welch's t-tests show significant differences between diabetic and non-diabetic patients in Glucose, BMI and Age.
- **Correlation and regression:** Insulin–Glucose $r = 0.4900$ is the strongest of the three pairs studied; Age explains only 7.23% of the variation in Glucose.

8. Conclusion

This project examined whether routine patient healthcare data can statistically predict diabetes in the Pima Indians Diabetes Dataset (768 patients, 9 variables). The evidence from every technique used points to the same conclusion.

- **Statistical analysis and probability:** 34.90% of patients are diabetic. Knowing that a patient's glucose exceeds 130 mg/dL raises the probability of diabetes to 63.64%, and Bayes' theorem gives a posterior of 75.00% for glucose above 150 mg/dL. Glucose therefore substantially improves the prediction of diabetes over the baseline prior.
- **Distribution analysis and CLT:** Glucose and Insulin are right-skewed and Age is clearly non-normal, so raw-data normality cannot be assumed. Because the sample sizes are large, the CLT makes inference on the mean valid, and the normal, binomial and exponential models give useful risk and planning probabilities.

- **Confidence intervals and hypothesis testing:** the confidence interval and the Z-tests show that glucose levels are significantly above the 120 mg/dL benchmark in the sampled groups, and Welch's t-tests show highly significant differences (all $p < 0.001$) between diabetic and non-diabetic patients in Glucose (+31.68 mg/dL), BMI (+4.55 kg/m²) and Age (+5.88 years).
- **Correlation and regression:** correlations with Outcome are all positive, with Glucose the strongest ($r = 0.4960$). Age has a statistically significant but weak effect on glucose (slope 0.6966 mg/dL per year, $R^2 = 7.23\%$).
- **Visualization:** the histograms, boxplot, heatmap, scatter plot matrix and density plots show clear separation of the diabetic and non-diabetic groups on Glucose and weaker separation on BMI and Age.

Major diabetes risk factors identified: Glucose is the primary risk factor; Insulin, BMI, SkinThickness, Age and Pregnancies are moderate risk factors; and BloodPressure and DiabetesPedigreeFunction are minor factors in this dataset. Healthcare organizations should prioritise glucose screening, weight management (BMI) and monitoring of older patients and of patients with elevated insulin.

Limitations and further work. The results are associations from an observational sample and do not prove causation; no single variable classifies patients perfectly because the groups overlap; and zero-valued measurements had to be treated as missing data. A multivariable predictive model such as logistic regression that combines Glucose, Insulin, BMI, Age and Pregnancies would be the natural next step to convert these statistical findings into a diabetes prediction tool.

9. References

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